BUSY OR NOT

Airport Security Screening Dashboard

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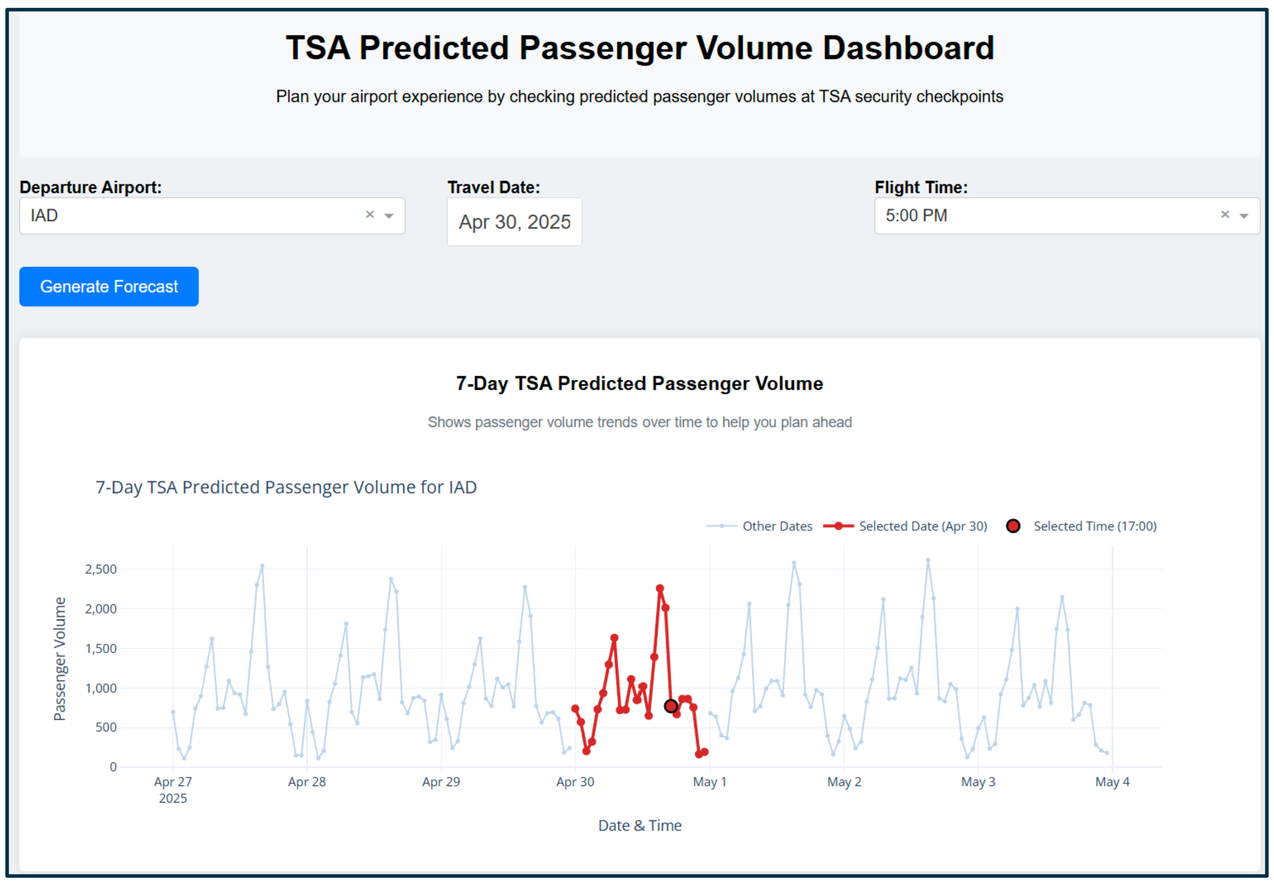
Motivation and Introduction

Millions of passengers pass through airport security checkpoints every week. Limited information makes predicting how busy the security screening line will be and knowing when to leave for the airport a challenge. This project utilizes hourly passenger throughput data published by the U.S. Transportation Security Administration (TSA) to model airport-specific passenger volume and display predictions in an interactive dashboard for travelers.



Users have the option to choose their airport, when they're traveling, and the estimated flight departure time. The model will provide predicted passengers.

Approaches

Predicted hourly number of travelers passing through a checkpoint are forecast using the gradient boosting regression algorithm XGBoost. Models are trained using recent historical data (excluding the COVID-19 outlier years of 2020-2021) for each airport with month, day, hour, day of week, and checkpoint as the inputs. Grid search cross validation is used for hyperparameter tuning to ensure optimal model performance. The model is then exported for quick future access. Models can be recreated when new data becomes available to ensure accurate predictions.

Users input the date and hour of flights they're considering as well as their primary airport, and the program will access the necessary models to provide throughput data for the selected airport, as well as other nearby options.

Existing approaches center around real-time security wait estimates. But these can become outdated by the time you arrive at the airport. Meanwhile, existing predictive products only provide airport-wide wait time estimates. At airports where each terminal has its own security checkpoint, these estimates provide little information. Existing predictive solutions also fail to consider holiday travel periods and do not simultaneously compare the throughput of multiple nearby airports.

Data

The data was sourced from 426 PDFs published on the TSA's FOIA Reading Room site. The PDFs contain hourly throughput data checkpoints at 455 airports across the U.S. from January 2017 to February 2025 – over 22.4 million rows in total. The data was extracted from the PDFs and converted into time series data.

We grouped airports based on location, size, and number of nearby airports to identify potential alternatives available to the user after they select their primary airport. To enable this, TSA throughput data is merged with a dataset from DataHub which includes latitude and longitude, and size classifications for U.S. airports. By combining these datasets, the system can recommend nearby alternatives and compare expected crowd levels at each location.

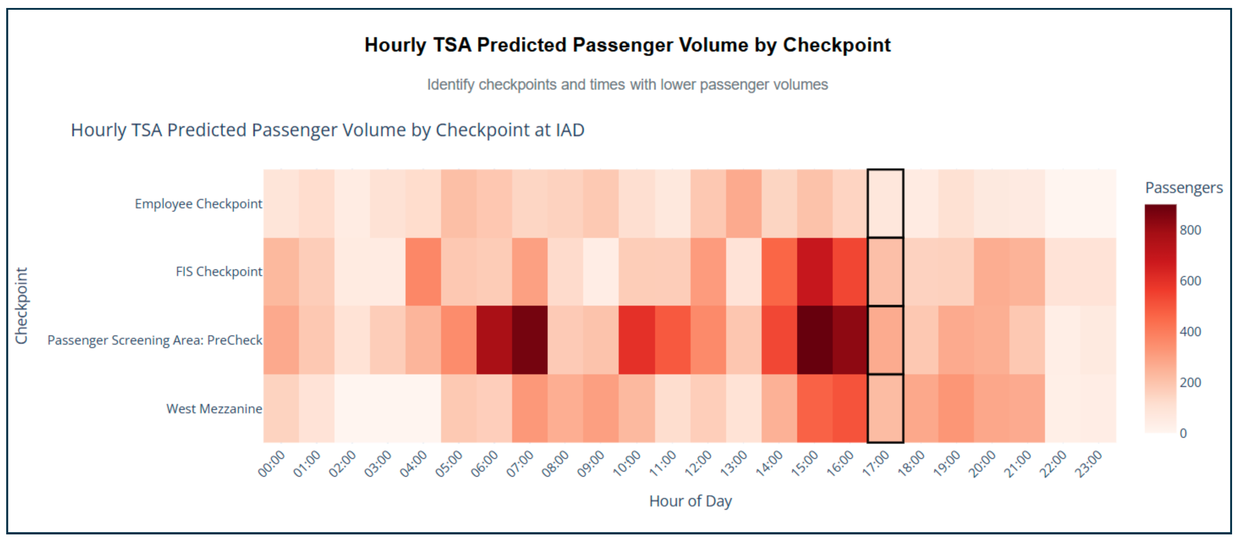
Experiments and Results

We divided the data into training and testing sets to assess the performance of ARIMA, linear regression, random forest, XGBoost, and a long short-term memory (LSTM) neural network models in predicting passenger throughput.

Root mean square error was the primary evaluation metric. R-squared was also used as an evaluation metric for the regression, random forest, and XGBoost models. Time series data for a single airport, Washington Dulles International Airport, was used to perform preliminary evaluation of the selected models. This airport was chosen due to its large passenger volume and multiple security checkpoints that operate out of a central check-in area.

The random forest model performed best in testing, but its computational cost was too high. XGBoost was slightly less accurate but could be trained about 600 times faster (0.2 seconds vs. about 2 minutes). Due to the strong overall performance and efficiency, XGBoost was selected.

We trained, tuned, and tested XGBoost models for all available airports. The R^2 scores stayed relatively stable across major airports in the range of 0.75 to 0.9. However, there were some outliers. Charlotte Douglas International Airport and Roanoke-Blacksburg Regional Airport were two outliers with R^2 scores of 0.577 and 0.377 respectively. This signaled that XGBoost may face challenges with airports that exhibit high seasonality or low traffic.



The dashboard includes a heat map of the projected passengers that will pass through each checkpoint at a selected airport on a given day.

Image Credits: Airport Flip Board Vectors by Vecteezy; Aiwygiterour via PxHere